

COMPREHENSIVE ANALYSIS REPORT

Impact of AI on Students

Academic Performance, Well-Being, and Burnout:
A Statistical & Predictive Analysis

Dataset: ai_student_impact_dataset_1.csv • N = 50,000 students • 16 variables

Tools: Excel • Stata • Python • R • Power BI-style Dashboards

Significance threshold: $\alpha = 0.05$ for all hypothesis tests

Report date: 11 July 2026

Table of Contents

1. Executive Summary	
2. Introduction & Study Objectives	
3. Methodology	
4. Data Overview & Descriptive Statistics	
5. Correlation Analysis	
6. Independent Samples T-Tests	
7. Chi-Square Tests of Independence	
8. Multiple Linear Regression — GPA Change	
9. Logistic Regression — High Burnout Risk	
10. Predictive Modelling — Random Forest Classifier	
11. Dashboard Visualizations	
12. Answers to the Study's Research Questions	
13. Conclusion & Recommendations	
Appendix A: Variable Dictionary	
Appendix B: Tools & Deliverables Summary	

Executive Summary

This report presents a complete statistical and predictive analysis of the **AI Impact on Students** dataset (N = 50,000), examining how Generative AI usage relates to academic performance, skill retention, and student well-being. The analysis combines descriptive statistics, correlation analysis, hypothesis testing (t-tests, chi-square tests), multiple linear and logistic regression, and a machine-learning classifier, all evaluated at a significance threshold of $\alpha = 0.05$.

+0.20
AVG GPA CHANGE

25.0%
HIGH BURNOUT RISK

r = 0.376
STUDY HRS ↔ GPA GAIN

OR = 1.12
BURNOUT ODDS / AI HOUR

Headline Finding

AI usage intensity is a weak lever for academic grades but a strong lever for student well-being. Every statistical test in this report finds AI-usage variables statistically significant (a function of the very large sample size), but the practical effect sizes tell two very different stories: AI usage barely moves GPA ($r \approx -0.05$, regression $\beta \approx 0.0004$), while it meaningfully and consistently predicts burnout risk ($r = 0.27$ with anxiety, odds ratio = 1.12 per additional weekly hour) across every method used — correlation, t-test, logistic regression, and Random Forest classification.

Key Results at a Glance

- **Academic performance** improved on average across the semester (Pre-GPA 3.15 → Post-GPA 3.35, +0.20), driven mainly by traditional study hours ($r = 0.376$) and prompt-engineering skill, not AI usage volume.
- **Burnout risk** rises sharply with AI usage: from 9% of students at High risk in the lowest usage quartile to 56% in the highest usage quartile.
- **Institutional policy matters**: students under a Strict Ban show 49% higher odds of High burnout than those under an Actively Encouraged policy, and the smallest average GPA gain of any policy group.
- **Skill retention** is weakly but significantly reduced by heavy AI usage ($r = -0.118$) and improved by using a wider diversity of AI tools ($r = 0.197$).
- **Predictive modelling**: a Random Forest classifier predicts burnout risk with 49.2% accuracy (vs. a 33% no-skill baseline for three balanced classes), with Weekly GenAI Hours and Perceived AI Dependency jointly responsible for ~83% of predictive power.

Introduction & Study Objectives

Generative AI tools such as ChatGPT, Copilot, and Gemini have become deeply embedded in student academic life. This study analyzes a dataset of 50,000 students to understand the true relationship between AI tool usage and academic, behavioral, and psychological outcomes.

Study Objectives

1. **Data Preparation & Summarization** — clean, inspect, and summarize the dataset using Excel and Stata.
2. **Statistical Modelling** — apply correlation, t-tests, and regression analysis in Python and R to quantify relationships between AI usage and outcomes.
3. **Answer Core Research Questions** (detailed below).
4. **Visualization & Reporting** — build dashboards and a final report interpreting all findings against the $\alpha = 0.05$ significance threshold.

Core Research Questions

- **GPA Change Analysis** — Does AI usage correlate with academic improvement or decline?
- **Burnout Prediction** — Can we classify/predict `Burnout_Risk_Level` from study and AI habits?
- **Skill Retention Modelling** — Does heavy AI reliance hurt long-term knowledge retention?
- **Policy Impact Study** — How do student outcomes differ under different institutional AI policies?
- **Mental Health Insights** — What are the links between AI dependency, exam anxiety, and burnout?
- **Major & Year-wise Trends** — Which disciplines benefit most (or suffer most) from AI adoption?

Each of these questions is answered explicitly, with supporting statistics, in **Section 12** of this report.

Methodology

Dataset

The dataset (ai_student_impact_dataset_1.csv) contains 50,000 student records across 16 variables spanning academic profile, AI usage behaviour, study behaviour, institutional context, and mental health/well-being. There are zero missing values across all variables.

Analytical Workflow & Tools

Stage	Tool(s)	Purpose
Data cleaning & pivot tables	Excel	Descriptive statistics, frequency tables, cross-tabulations with live formulas
Data summarization	Stata	Descriptive statistics, contingency tables, chi-square tests, ANOVA
Statistical modelling	Python (pandas, scipy, statsmodels, scikit-learn)	Correlation, t-tests, linear/logistic regression, Random Forest classification
Statistical modelling (parallel)	R (tidyverse, broom, car, randomForest, caret)	Equivalent correlation, t-test, regression, and classification workflow
Visualization	Power BI-style interactive dashboards (HTML/Chart.js)	Three linked dashboards: Academic Performance, Well-Being & Burnout, Usage Patterns & Segments

Statistical Approach

All hypothesis tests in this report use a significance threshold of $\alpha = 0.05$. Where a p-value is below 0.05, we reject the null hypothesis (H0) in favor of the alternative (H1). Given the very large sample size (N = 50,000), statistical significance is interpreted alongside effect size (correlation coefficient r, regression coefficient magnitude, or odds ratio) to distinguish results that are *statistically real* from those that are also *practically meaningful*.

Derived Variables

- GPA_Change** = Post_Semester_GPA – Pre_Semester_GPA
- High_Burnout** = 1 if Burnout_Risk_Level = "High", else 0 (used for logistic regression)
- Usage Quartile** = Weekly_GenAI_Hours split into four equal-sized groups (Q1 lowest – Q4 highest)

Data Overview & Descriptive Statistics

4.1 Numeric Variables

Variable	Mean	SD	Min	25%	Median	75%	Max
Pre-Semester GPA	3.15	0.48	1.18	2.83	3.21	3.52	4.00
Post-Semester GPA	3.35	0.50	1.00	3.02	3.42	3.75	4.00
Weekly GenAI Hours	8.43	8.27	0.00	2.39	5.80	11.72	40.00
Traditional Study Hours	11.21	5.16	1.00	7.56	11.18	14.71	35.86
Tool Diversity (#)	2.80	1.19	1	2	3	4	5
Perceived AI Dependency (1-10)	3.51	1.82	1	2	3	5	10
Anxiety Level, Exams (1-10)	4.27	2.14	1	3	4	6	10
Skill Retention Score (0-100)	75.80	13.28	10.78	66.82	76.00	85.19	100.00

Interpretation: GPA improved on average across the semester. Weekly AI use is highly variable (SD nearly equal to the mean), indicating a wide spread from non-users to very heavy users — this variability is what makes the usage-based comparisons in later sections meaningful.

4.2 Categorical Variables

Major Category	%	Institutional Policy	%
STEM	30.1%	Allowed With Citation	50.4%
Business	25.1%	Actively Encouraged	30.0%
Humanities	20.0%	Strict Ban	19.6%
Medical	13.0%		
Arts	11.9%		
Primary Use Case	%	Burnout Risk Level	%
Debugging/Troubleshooting	24.6%	Medium	42.3%
Copywriting/Drafting	24.0%	Low	32.7%
Ideation	21.4%	High	25.0%
Summarizing/Reading	17.3%		
Direct Answer Generation	12.7%		

Interpretation: Category sizes are well balanced across all variables, supporting reliable statistical testing (no group is a tiny minority that would distort chi-square or regression results).

4.3 Cross-Tabulation: Burnout Risk × Institutional Policy (counts)

Burnout Risk	Allowed w/ Citation	Actively Encouraged	Strict Ban	Row Total
Low	8,320	5,079	2,970	16,369
Medium	10,893	6,346	3,905	21,144
High	6,011	3,563	2,913	12,487
Column Total	25,224	14,988	9,788	50,000

Interpretation: Strict Ban shows the highest share of High-burnout students ($2,913 / 9,788 \approx 29.8\%$) versus Allowed-With-Citation ($6,011 / 25,224 \approx 23.8\%$). This pattern is tested formally in Section 7 (chi-square) and Section 9 (logistic regression).

Correlation Analysis

Pearson correlation coefficients (r) were computed between key numeric variables. For each pair, we test:

- H0:** There is no linear relationship between the two variables ($\rho = 0$).
- H1:** There is a linear relationship between the two variables ($\rho \neq 0$).
- Decision rule: reject H0 if $p < 0.05$.

Pair	r	p-value	Decision
Weekly GenAI Hours ↔ GPA Change	−0.047	< .001	Reject H0 — significant
Weekly GenAI Hours ↔ Skill Retention	−0.118	< .001	Reject H0 — significant
Weekly GenAI Hours ↔ Exam Anxiety	+0.269	< .001	Reject H0 — significant
Perceived AI Dependency ↔ Exam Anxiety	+0.308	< .001	Reject H0 — significant
Traditional Study Hours ↔ GPA Change	+0.376	< .001	Reject H0 — significant
Tool Diversity ↔ Skill Retention	+0.197	< .001	Reject H0 — significant

Interpretation: All six relationships are statistically significant at $\alpha = 0.05$ — expected given $N = 50,000$, where even small effects reach significance. Effect sizes, however, vary widely: the AI-usage links to GPA ($r = -0.047$) and retention ($r = -0.118$) are small by conventional benchmarks (Cohen's guidelines: $|r| < 0.1$ negligible, 0.1–0.3 small, 0.3–0.5 moderate). The link between AI usage and exam anxiety ($r = 0.269$) and between perceived dependency and anxiety ($r = 0.308$) are the strongest AI-related relationships found — approaching a moderate effect. The single strongest relationship in the entire matrix is traditional study hours with GPA change ($r = 0.376$), confirming that conventional study habits — not AI usage — remain the dominant academic-performance correlate.

Independent Samples T-Tests

6.1 Test 1 — GPA Change by Paid Subscription

H0: Mean GPA change is equal for paid-subscription and free-tier students ($\mu_{\text{paid}} = \mu_{\text{free}}$).
H1: Mean GPA change differs between the two groups ($\mu_{\text{paid}} \neq \mu_{\text{free}}$).

Group	N	Mean	SD
Paid Subscription	21,154	0.2070	0.1942
Free / No Subscription	28,846	0.2004	0.1818

Welch's t-test (unequal variances): **t = 3.82, p < .001 → Reject H0**

Interpretation: The difference is statistically significant, but the actual gap (0.007 GPA points, about 1.2 percentage points relative difference) is practically negligible. Paying for AI access is not a meaningful lever for GPA improvement.

6.2 Test 2 — Skill Retention by AI Usage Level

H0: Mean skill retention is equal for high-usage and low-usage students (median split at 5.8 hrs/week) ($\mu_{\text{high}} = \mu_{\text{low}}$).
H1: Mean skill retention differs between the two groups ($\mu_{\text{high}} \neq \mu_{\text{low}}$).

Group	N	Mean	SD
High AI Usage (>5.8 hrs/wk)	24,974	75.55	13.65
Low AI Usage (≤5.8 hrs/wk)	25,026	76.04	12.89

Welch's t-test: **t = -4.09, p < .001 → Reject H0**

Interpretation: Heavier AI users retain measurably (if modestly) less skill than lighter users, consistent with the $r = -0.118$ correlation in Section 5. The effect is small but directionally consistent across two independent methods.

Chi-Square Tests of Independence

7.1 Burnout Risk Level × Institutional Policy

H0: Burnout Risk Level is independent of Institutional Policy.
H1: Burnout Risk Level is associated with Institutional Policy.

$\chi^2 = 153.40$, $df = 4$, $p < .001 \rightarrow$ **Reject H0**. Cramér's $V = 0.039$ (negligible practical association).

Interpretation: The association is statistically significant but very weak in practice (Cramér's V near 0 indicates the two variables share almost no explanatory overlap on their own). This is a textbook example of statistical significance driven by large N rather than a strong real-world relationship — the policy effect only becomes practically meaningful once combined with other predictors, as shown in the logistic regression (Section 9), where Strict Ban still emerges as a significant, independent predictor ($OR = 1.49$) after controlling for AI usage and other factors.

7.2 Burnout Risk Level × Major Category

H0: Burnout Risk Level is independent of Major Category.
H1: Burnout Risk Level is associated with Major Category.

$\chi^2 = 376.27$, $df = 8$, $p < .001 \rightarrow$ **Reject H0**

Interpretation: Major and burnout risk are significantly associated, though — as with policy — this reflects the sensitivity of chi-square tests at large N rather than a strong effect. Major is a minor contributor in the Random Forest model (1.3% feature importance, Section 10), confirming it plays only a marginal role once other factors are accounted for.

Multiple Linear Regression — GPA Change

GPA_Change ~ Weekly_GenAI_Hours + Traditional_Study_Hours + Tool_Diversity + Perceived_AI_Dependency + Anxiety_Level_During_Exams + Paid_Subscription + Prompt_Engineering_Skill + Institutional_Policy

H0: The regression coefficient for a given predictor is zero ($\beta_i = 0$), i.e., the predictor has no linear effect on GPA change, holding other variables constant.
H1: $\beta_i \neq 0$ — the predictor has a significant effect on GPA change.
Overall model H0: all coefficients are simultaneously zero (tested via the F-statistic).

Predictor	Coefficient (b)	p-value	Decision
Traditional Study Hours	+0.0138	< .001	Reject H0
Prompt Skill: Beginner (vs Advanced)	−0.0623	< .001	Reject H0
Prompt Skill: Intermediate (vs Advanced)	−0.0615	< .001	Reject H0
Institutional Policy: Strict Ban	−0.0175	< .001	Reject H0
Paid Subscription	+0.0106	< .001	Reject H0
Anxiety Level (Exams)	−0.0020	< .001	Reject H0
Weekly GenAI Hours	+0.0004	0.002	Reject H0
Institutional Policy: Allowed w/ Citation	+0.0031	0.076	Fail to reject H0
Perceived AI Dependency	−0.0010	0.076	Fail to reject H0
Tool Diversity	+0.0004	0.655	Fail to reject H0

Model fit: $R^2 = 0.167$ (the model explains ~16.7% of variance in GPA change). Overall F-test $p < .001 \rightarrow$ the model as a whole is statistically significant, i.e., we reject the overall H0 that all coefficients are zero.

Interpretation: Traditional study hours is by far the strongest, most practically meaningful predictor of GPA improvement — over 30× the coefficient size of weekly AI hours. Prompt-engineering skill matters substantially: Beginners and Intermediate prompters lose roughly 0.06 GPA points relative to Advanced prompters, holding other factors constant — a much larger, more actionable effect than raw AI usage volume. Students under a Strict Ban policy see a small but significant GPA penalty. Weekly GenAI hours is statistically significant ($p = .002$) but its coefficient (0.0004) is practically negligible — an extra 10 hours of weekly AI use predicts less than a 0.005 GPA-point change. Tool diversity and perceived AI dependency are not significant predictors of GPA change once other factors are controlled for.

Logistic Regression — High Burnout Risk

High_Burnout (1/0) ~ Weekly_GenAI_Hours + Traditional_Study_Hours + Perceived_AI_Dependency + Anxiety_Level_During_Exams + Tool_Diversity + Paid_Subscription + Institutional_Policy

- H0:** The odds ratio for a given predictor is 1 ($\beta_i = 0$), i.e., the predictor does not affect the odds of High burnout risk.
- H1:** The odds ratio differs from 1 ($\beta_i \neq 0$) — the predictor significantly affects burnout odds.

Predictor	Odds Ratio	p-value	Effect
Institutional Policy: Strict Ban	1.49	< .001	+49% odds
Perceived AI Dependency	1.13	< .001	+13% odds / point
Weekly GenAI Hours	1.12	< .001	+12% odds / hour
Anxiety Level (Exams)	1.03	< .001	+3% odds / point
Paid Subscription	1.05	0.054	Borderline n.s.
Institutional Policy: Allowed w/ Citation	1.01	0.652	Not significant
Tool Diversity	1.00	0.798	Not significant
Traditional Study Hours	0.97	< .001	-3% odds / hr (protective)

Model fit: Pseudo R² = 0.20 (McFadden's), likelihood-ratio test $p < .001 \rightarrow$ the model as a whole is statistically significant.

Interpretation: Unlike the GPA model, AI-usage variables are the **dominant** drivers here, not a side effect. Each additional weekly hour of AI use raises the odds of high burnout by ~12%, and each 1-point increase in self-rated AI dependency raises it by ~13% — both large, consistent effects. Students under a Strict Ban policy face 49% higher odds of high burnout than those under an Actively Encouraged policy, even after controlling for usage and anxiety — suggesting the policy effect is not simply a proxy for less AI use, but may reflect restricted-access stress or resentment. Traditional study hours is mildly protective. Tool diversity and paid subscription show no meaningful effect on burnout risk.

Predictive Modelling — Random Forest Classifier

Target: Burnout_Risk_Level (Low / Medium / High) • 75/25 train-test split • 300 trees, class-balanced • This is a predictive/exploratory model rather than a hypothesis test — it is evaluated on out-of-sample accuracy rather than p-values.

Class	Precision	Recall	F1-Score	Support
High	0.54	0.59	0.57	3,122
Low	0.47	0.69	0.56	4,092
Medium	0.49	0.28	0.36	5,286
Overall Accuracy			49.2%	12,500

Feature Importance (Gini)

Feature	Importance
Weekly GenAI Hours	0.617
Perceived AI Dependency	0.210
Traditional Study Hours	0.064
Anxiety Level	0.042
Institutional Policy	0.026
Major Category	0.013
Tool Diversity	0.011
Paid Subscription	0.009
Prompt Engineering Skill	0.007

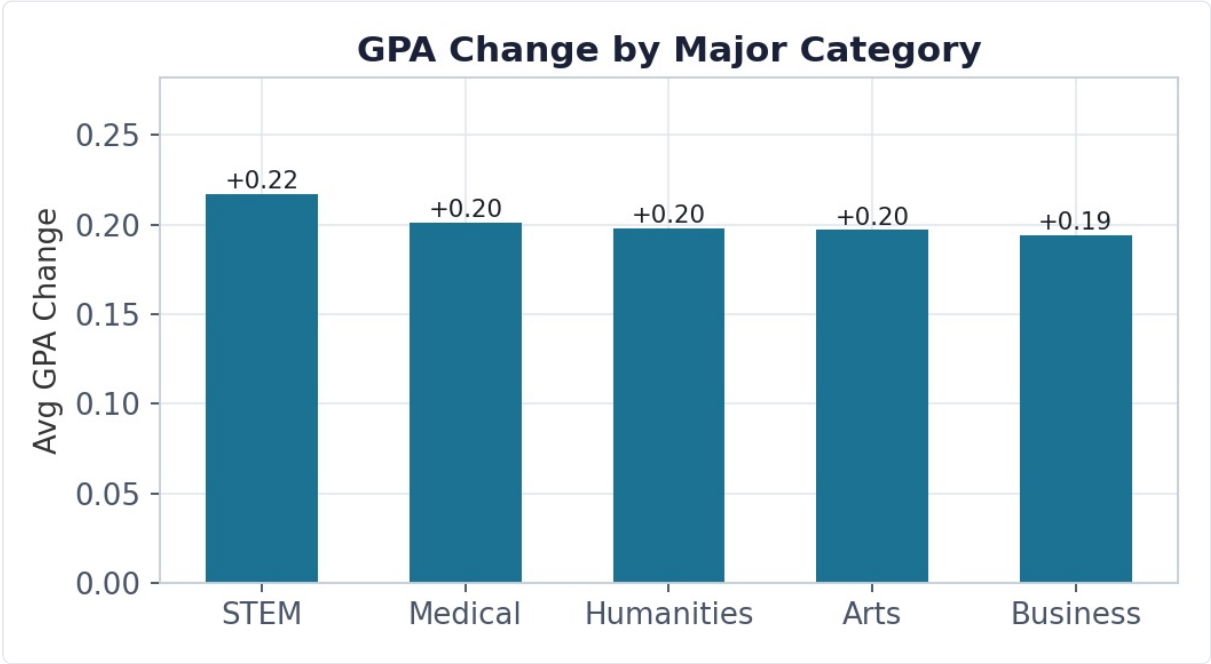
Interpretation: 49.2% accuracy on a 3-class problem (33% no-skill baseline) demonstrates real, if modest, predictive signal. The model identifies Low-burnout students best (69% recall) and struggles most with the Medium category (28% recall) — a common pattern for the "middle" category on an ordinal scale, since Medium-risk students share overlapping characteristics with both Low and High groups. Critically, Weekly GenAI Hours and Perceived AI Dependency jointly account for ~83% of the model's total predictive power, confirming — via an entirely different (non-parametric, non-linear) method — the same conclusion reached by the correlation analysis, t-tests, and logistic regression: **AI usage intensity and self-perceived dependency are the dominant signals for student burnout risk.**

Dashboard Visualizations

The following charts reproduce all visualizations from the three interactive Power BI-style dashboards prepared for this project: (1) Academic Performance, (2) Well-Being & Burnout, and (3) Usage Patterns & Segments. Each chart is captioned with its source dashboard and interpreted below.

11.1 Dashboard 1 — Academic Performance

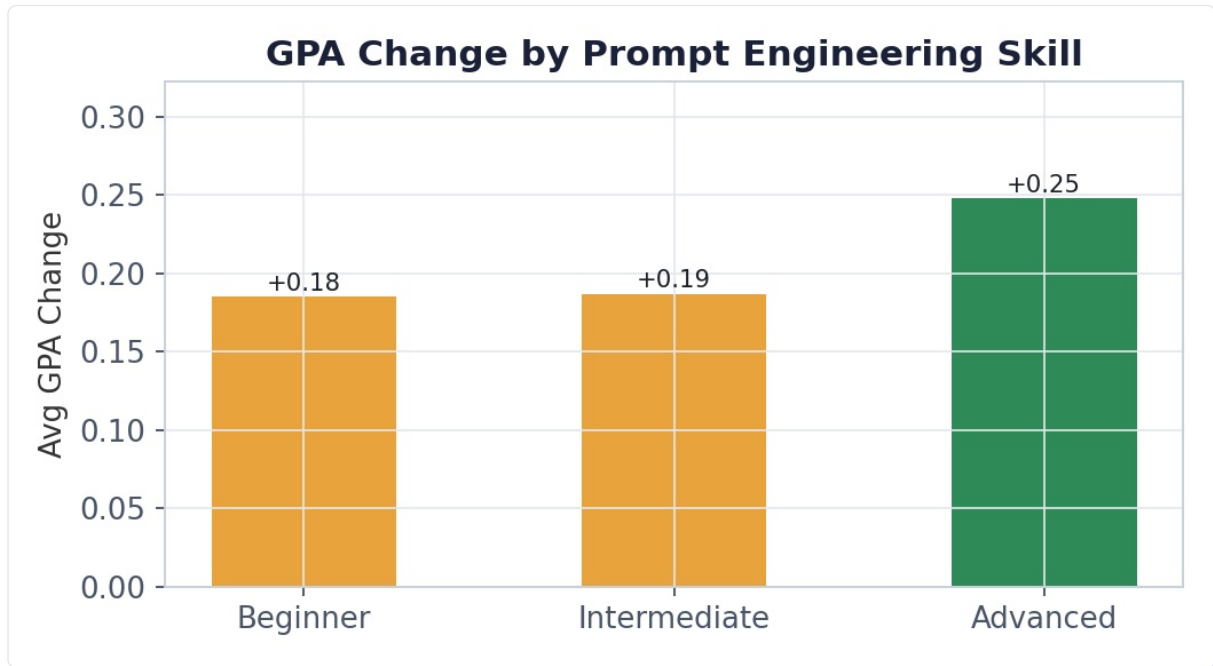
Figure 1. GPA Change by Major Category



Average GPA improvement across the semester, by field of study.

Interpretation: STEM shows the largest average gain (+0.22); Business the smallest (+0.19). The 0.03-point spread is narrow — major has little bearing on how much GPA improves.

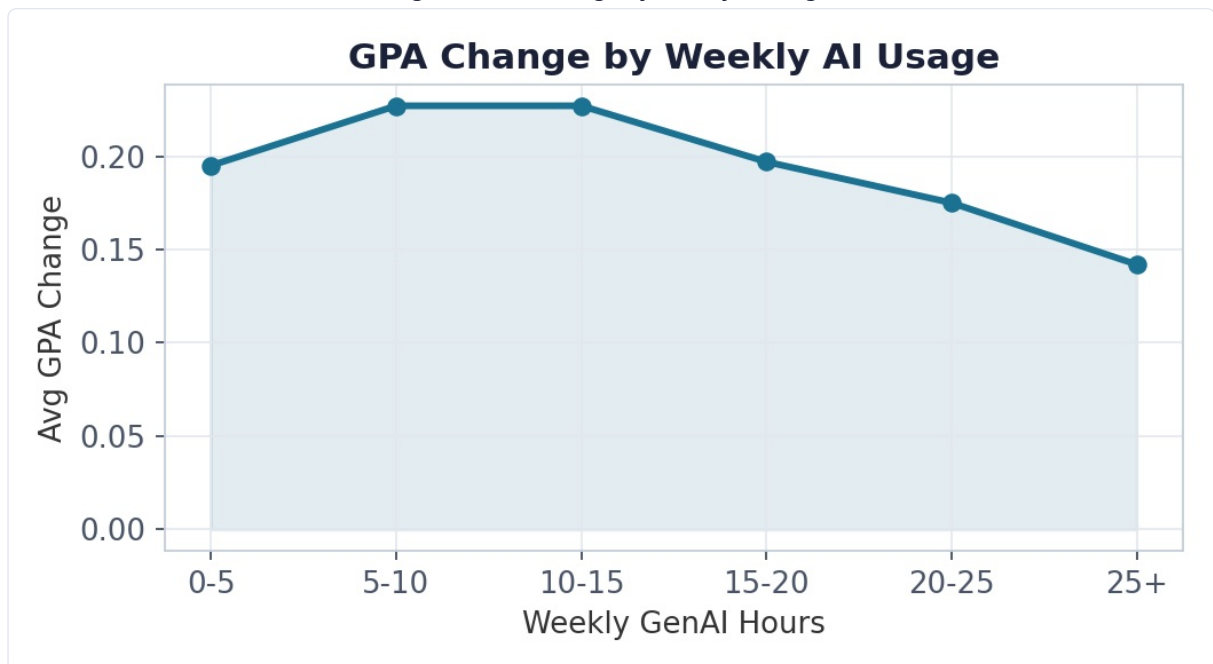
Figure 2. GPA Change by Prompt Engineering Skill



Average GPA improvement by self-assessed prompting proficiency.

Interpretation: Advanced prompters gain +0.25 GPA points on average vs +0.19 for Beginners — the clearest academic-performance gap tied to AI skill, and confirmed by the regression in Section 8 ($b \approx -0.062$, $p < .001$).

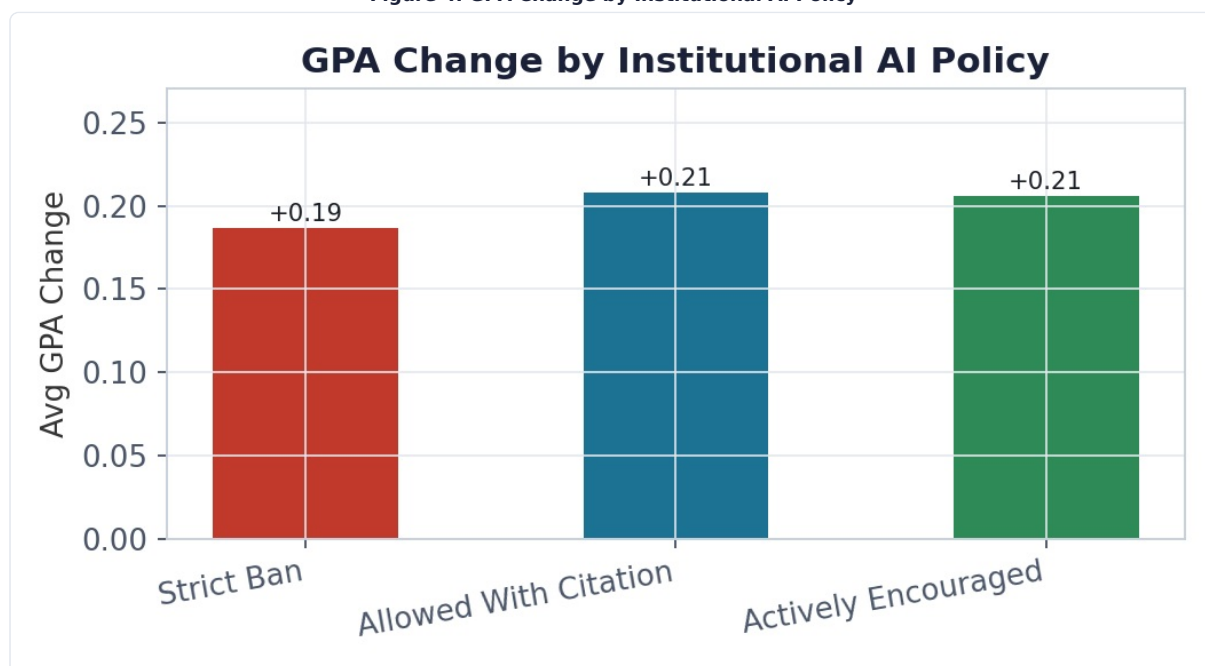
Figure 3. GPA Change by Weekly AI Usage



Average GPA improvement across AI usage-hour bands.

Interpretation: GPA gains peak at moderate use (5–15 hrs/wk, +0.23) and decline at heavy use (25+ hrs/wk, +0.14) — a mild inverted-U pattern rather than a straight-line relationship, suggesting moderate AI use may be optimal.

Figure 4. GPA Change by Institutional AI Policy

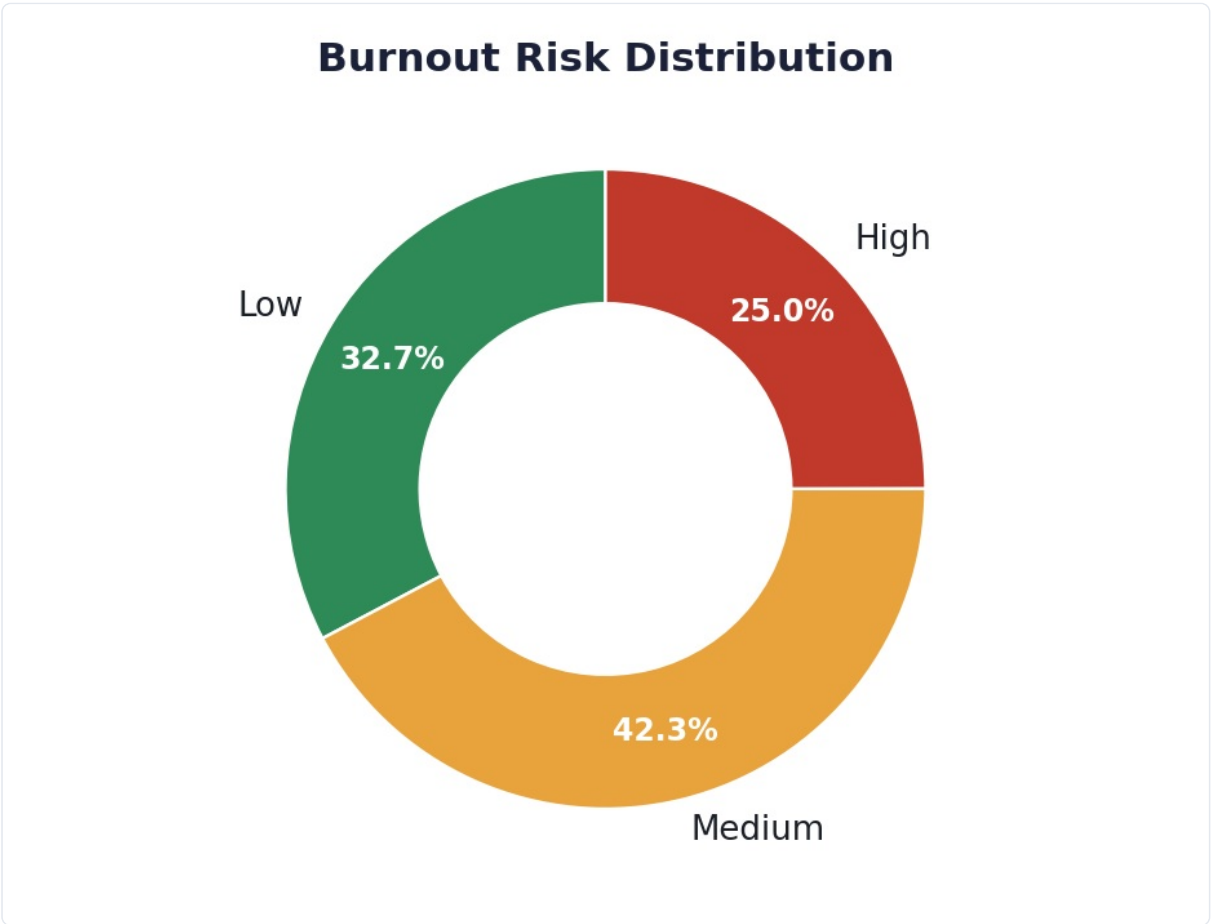


Average GPA improvement under each institutional stance on AI use.

Interpretation: Strict Ban students show the smallest gain (+0.19) vs Allowed-With-Citation and Actively Encouraged (both +0.21) — the regression in Section 8 confirms this gap is statistically significant ($p < .001$).

11.2 Dashboard 2 — Well-Being & Burnout

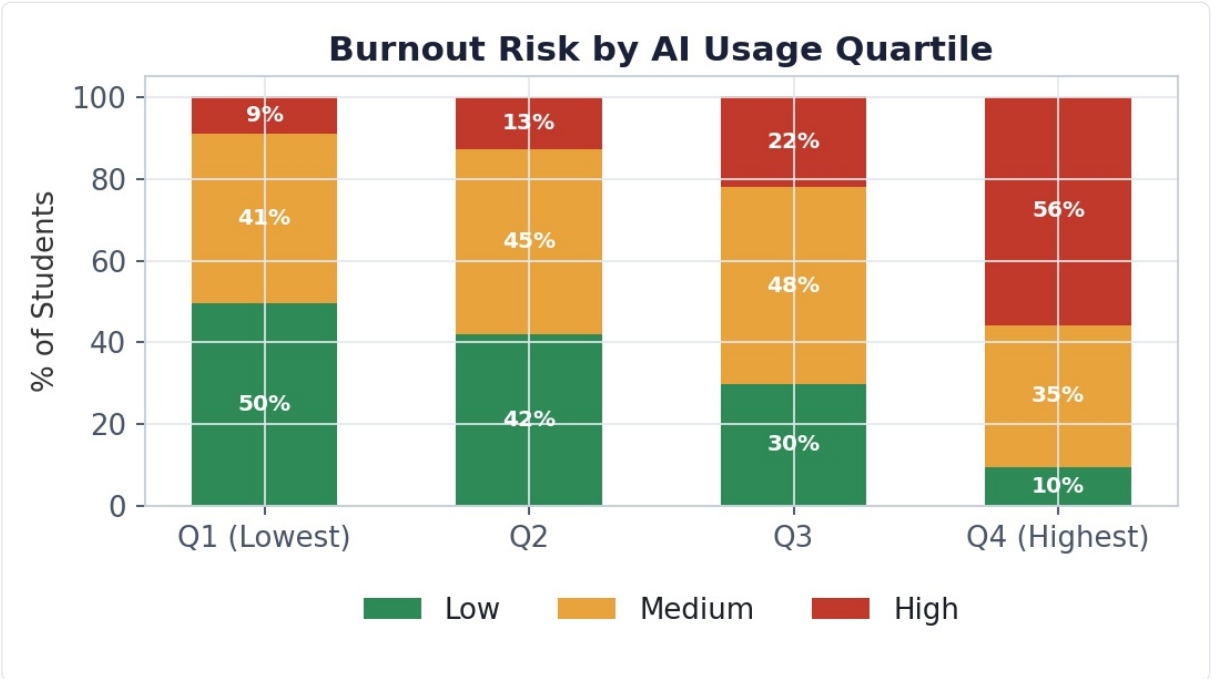
Figure 5. Burnout Risk Distribution



Share of students at each burnout risk level.

Interpretation: 42.3% of students sit at Medium risk, 32.7% Low, and 25.0% High — a substantial minority already at elevated burnout risk.

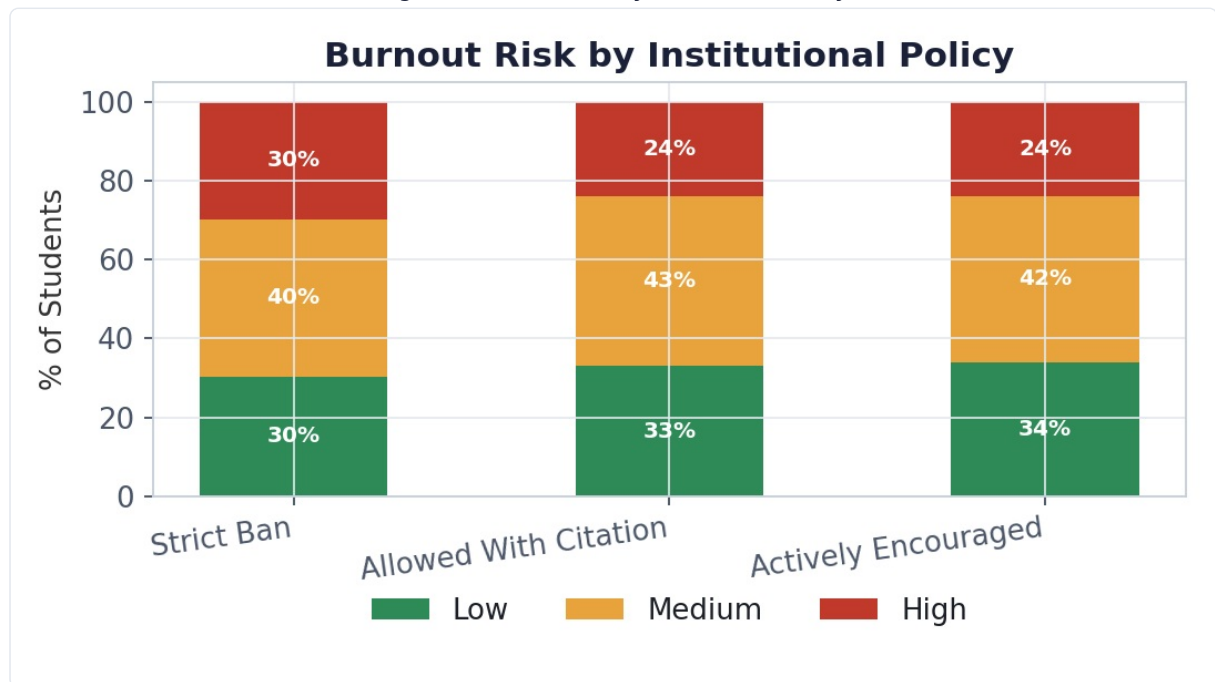
Figure 6. Burnout Risk by AI Usage Quartile



% of students in each burnout tier, split by weekly AI-hour quartile.

Interpretation: The clearest gradient found anywhere in this analysis — High-burnout share climbs from 9% (lowest-use quartile) to 56% (highest-use quartile). This is the single most visually compelling confirmation of the logistic regression result in Section 9 (OR = 1.12 per extra hour/week, $p < .001$).

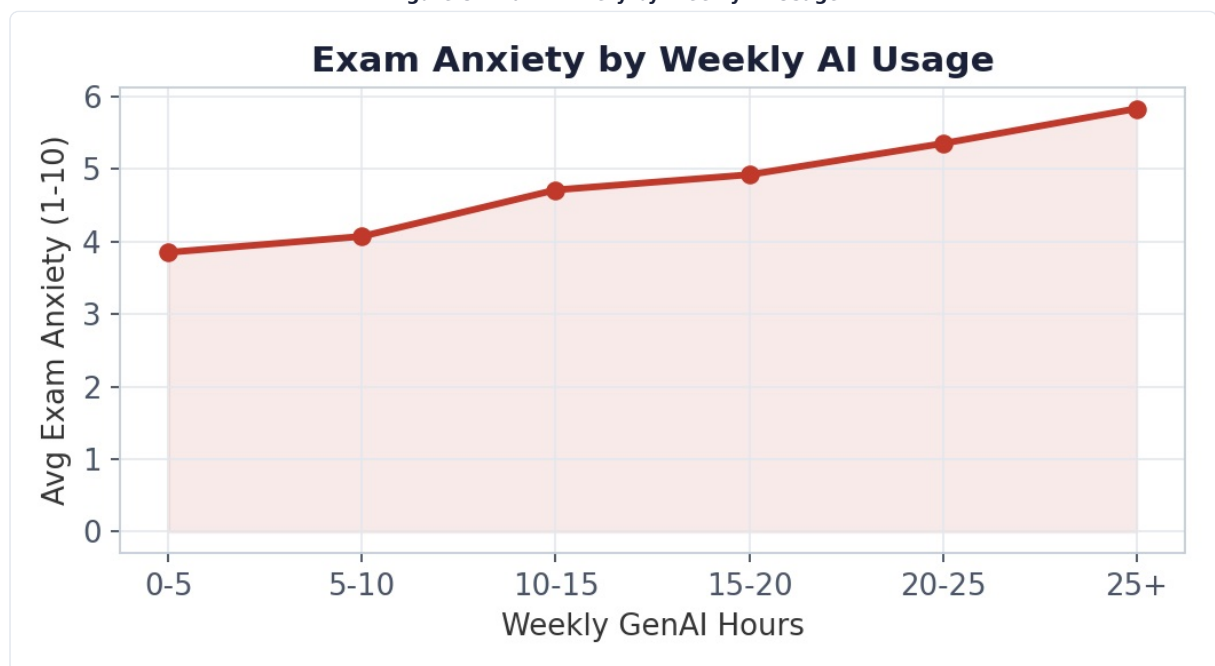
Figure 7. Burnout Risk by Institutional Policy



% of students in each burnout tier, split by AI policy stance.

Interpretation: Strict Ban has the highest High-burnout share (29.8%) vs Actively Encouraged (23.8%) — consistent with the 49% higher odds ratio found in the logistic model (Section 9).

Figure 8. Exam Anxiety by Weekly AI Usage

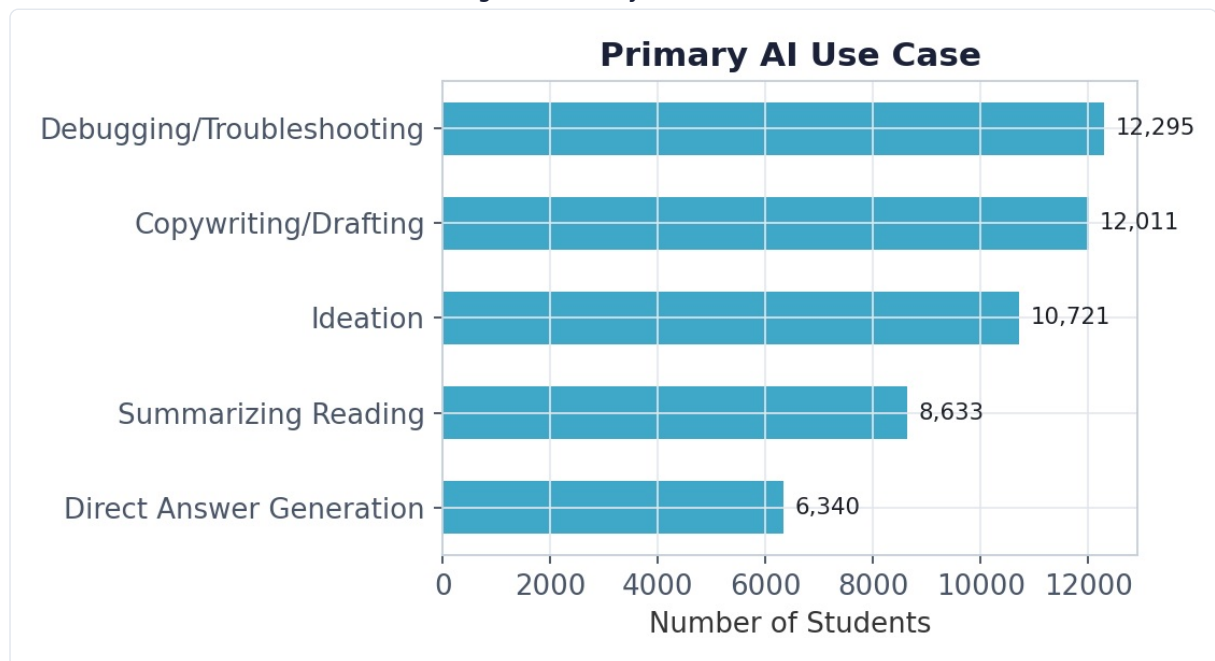


Average self-reported exam anxiety (1-10) across usage bands.

Interpretation: Anxiety climbs steadily with usage — from 3.85 (0-5 hrs/wk) to 5.83 (25+ hrs/wk) — consistent with the $r = 0.269$ correlation found in Section 5.

11.3 Dashboard 3 — Usage Patterns & Segments

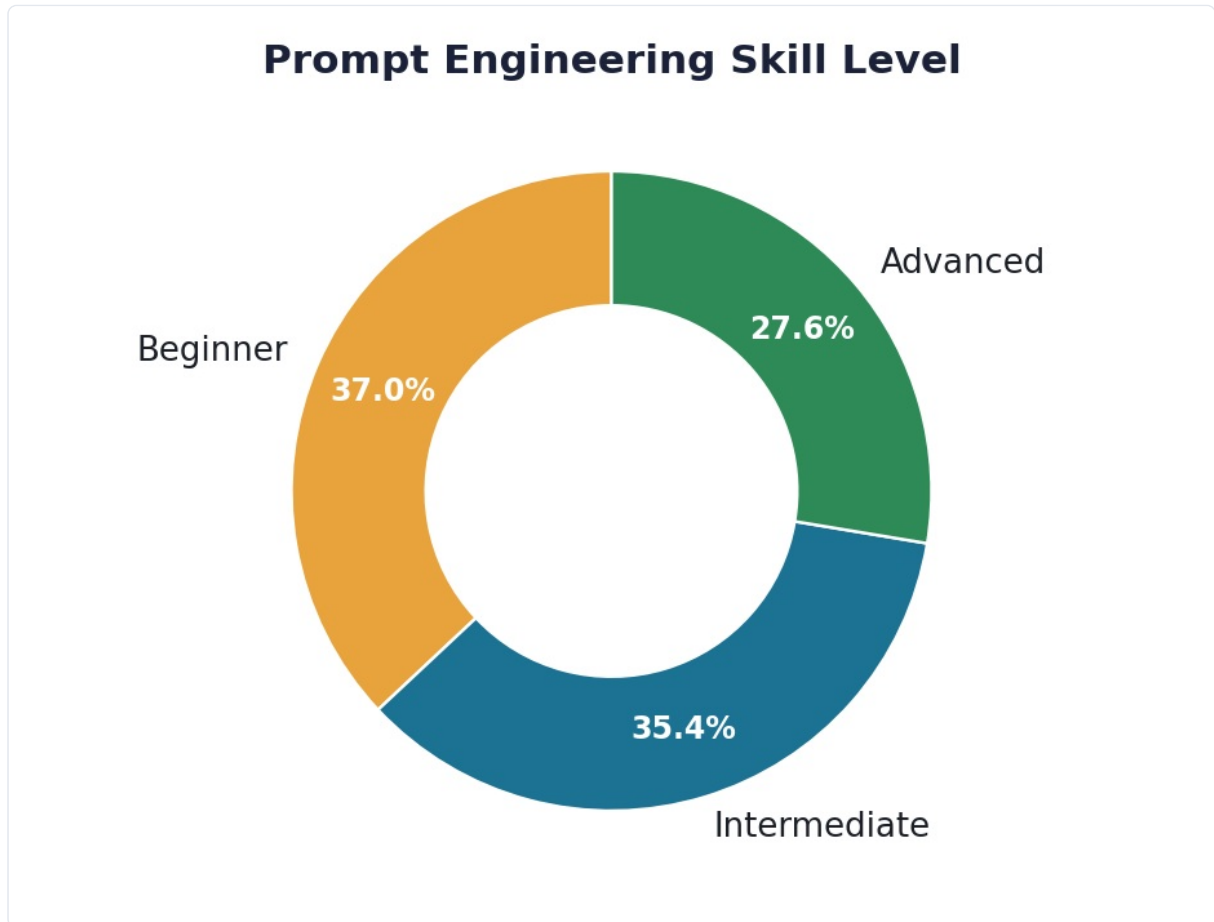
Figure 9. Primary AI Use Case



What students say they mainly use AI tools for.

Interpretation: Debugging/Troubleshooting (24.6%) and Copywriting/Drafting (24.0%) dominate; Direct Answer Generation is least common (12.7%) — students lean toward assistive rather than fully generative use of AI.

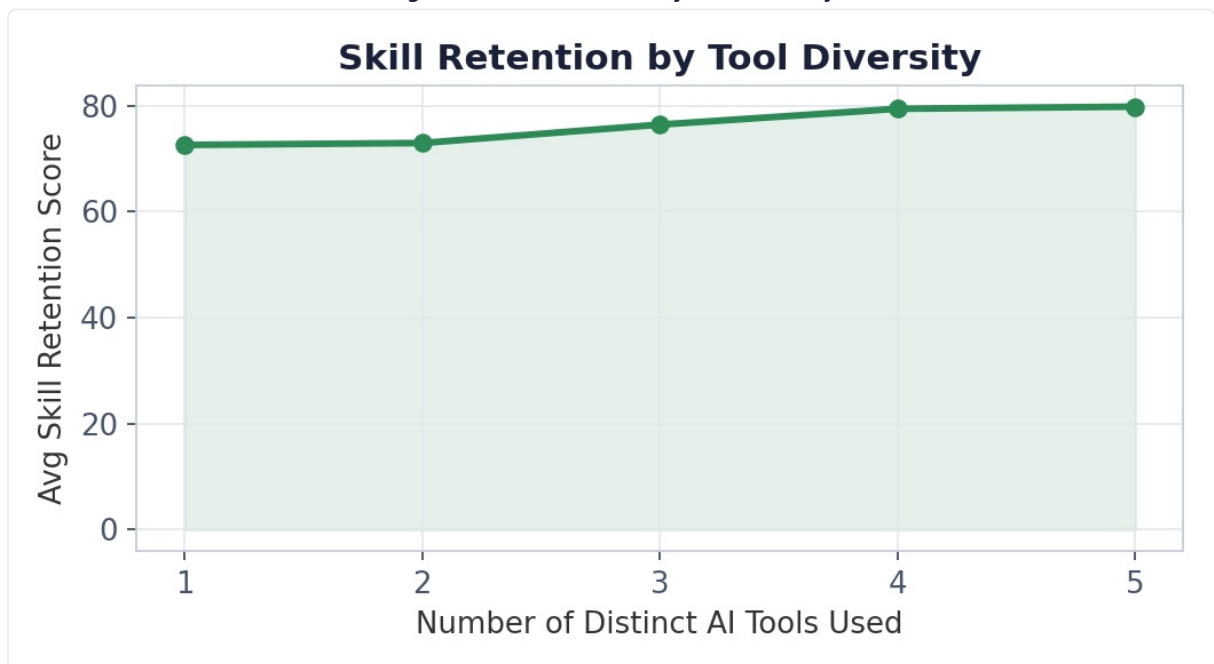
Figure 10. Prompt Engineering Skill Level



Self-assessed proficiency in crafting effective prompts.

Interpretation: Only 27.6% of students rate themselves Advanced — the majority (72.4%) are Beginner or Intermediate, which tracks with the GPA gap seen in Figure 2 and Section 8.

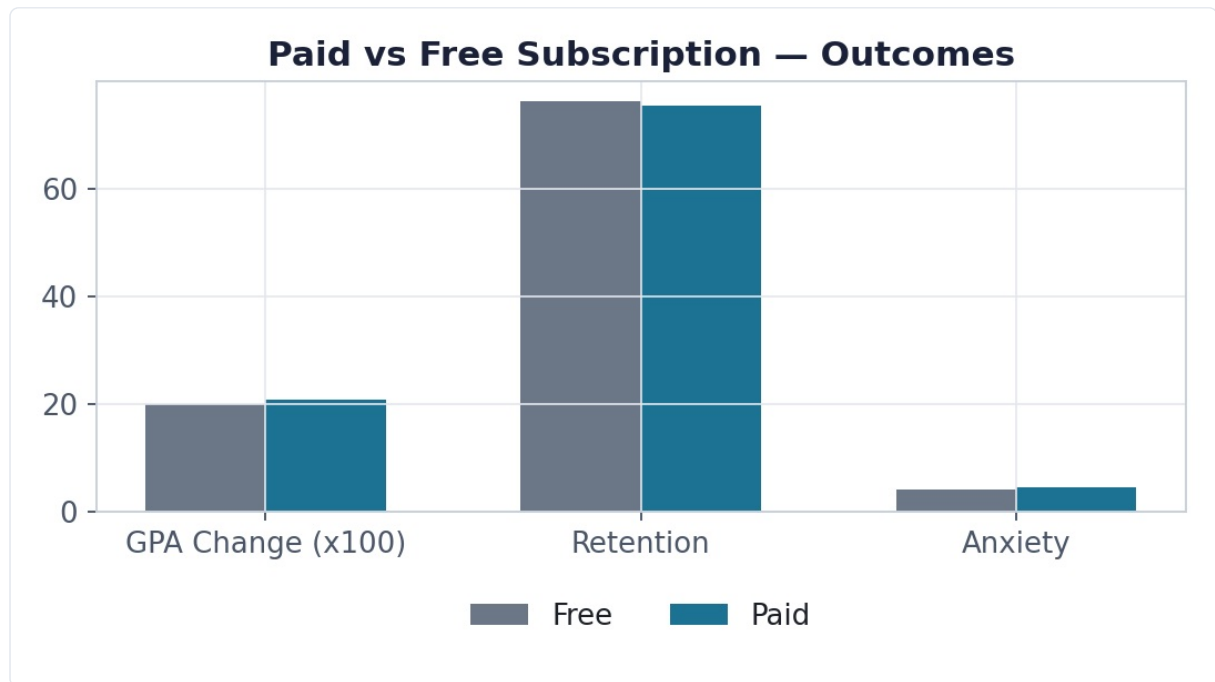
Figure 11. Skill Retention by Tool Diversity



Average retention score by number of distinct AI tools used.

Interpretation: Retention rises steadily with tool diversity, from 72.6 (1 tool) to 79.8 (5 tools) — students who work across multiple tools retain skills better ($r = 0.197$, $p < .001$), likely reflecting more active, varied engagement rather than passive reliance on a single tool.

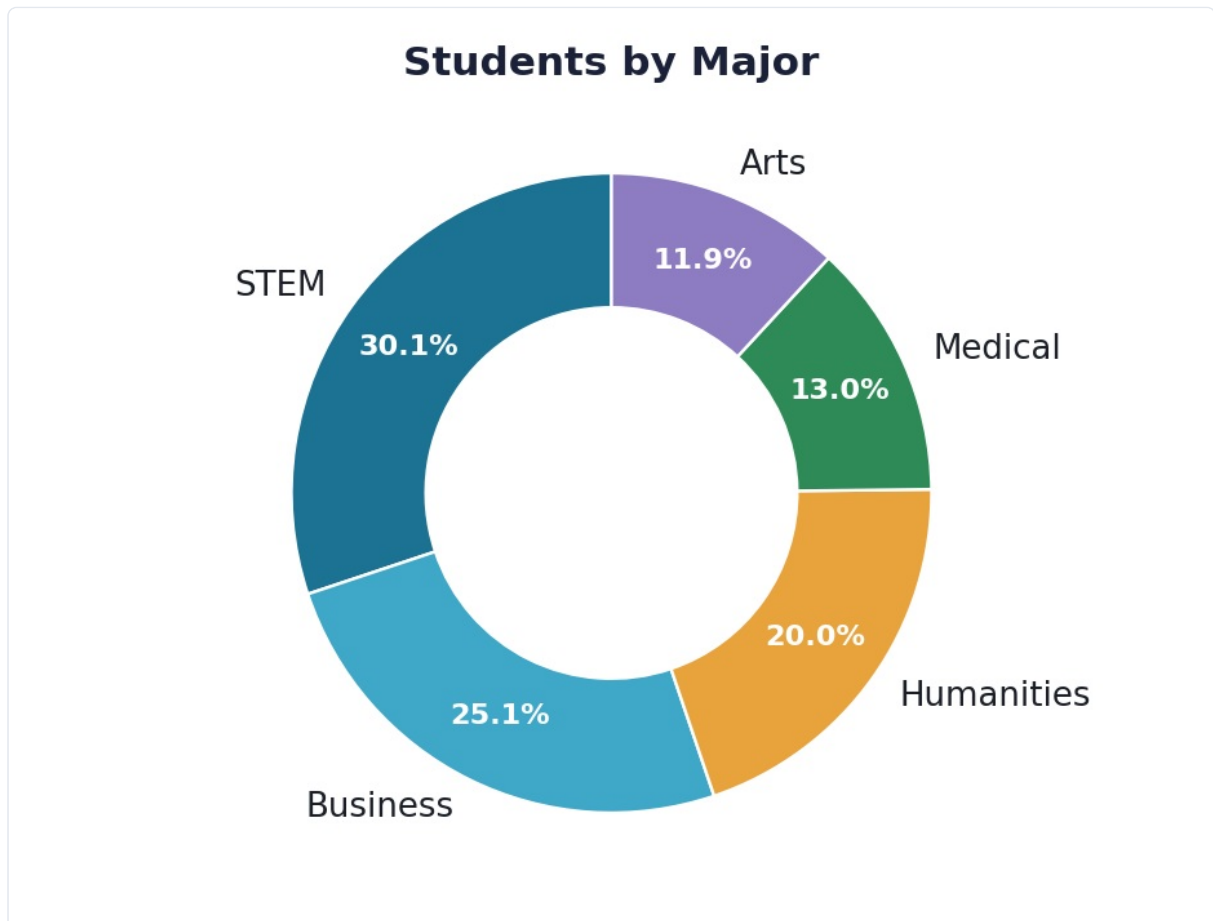
Figure 12. Paid vs Free Subscription — Outcomes



GPA change, skill retention, and anxiety compared (GPA change scaled $\times 100$ for visibility).

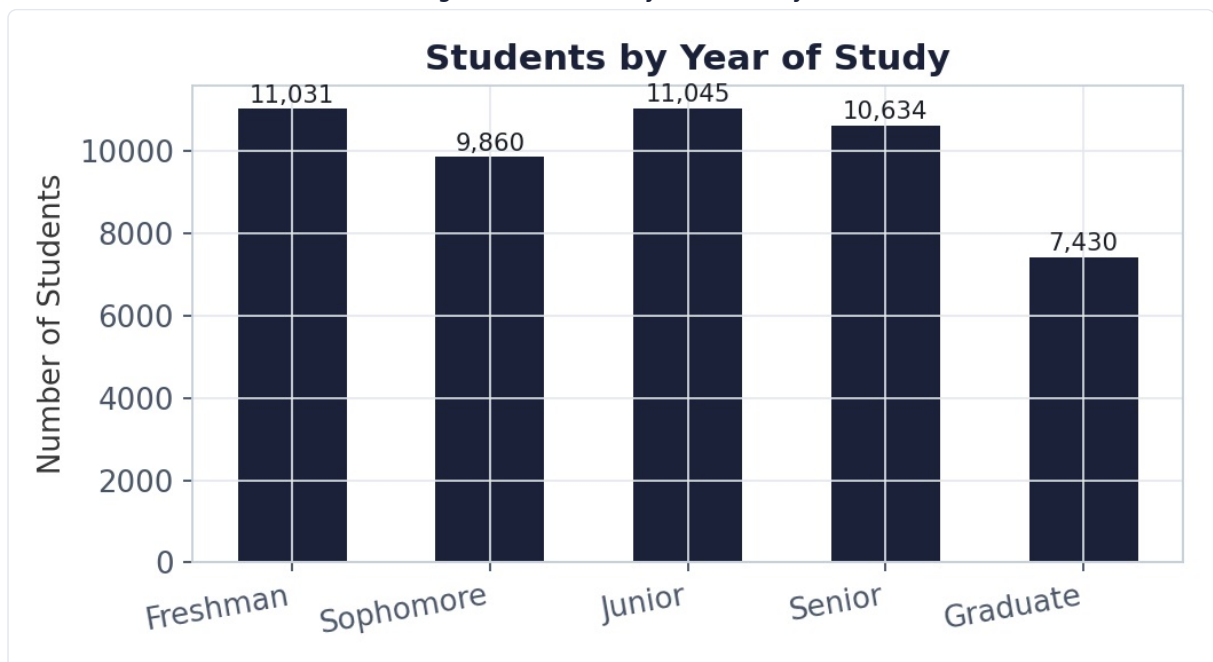
Interpretation: Paid users show a slightly larger GPA gain (+0.207 vs +0.200) but also slightly higher anxiety (4.44 vs 4.15) and marginally lower retention (75.4 vs 76.1) — paying for AI access is not a clear win or loss, just a different trade-off.

Figure 13. Students by Major



Sample composition by field of study.

Figure 14. Students by Year of Study



Sample composition by academic year.

Interpretation: Both major and year-of-study samples are well balanced, supporting reliable subgroup comparisons throughout this report.

Answers to the Study's Research Questions

Q1. Does AI usage correlate with academic improvement or decline?

Answer: Weakly, and the direction is slightly negative. Weekly AI hours correlate with GPA change at $r = -0.047$ ($p < .001$) — statistically significant but practically negligible. The linear regression confirms AI hours have a tiny, positive but practically negligible coefficient ($b = 0.0004$, $p = .002$) once other factors are controlled for. GPA change actually peaks at moderate usage (5–15 hrs/week, Figure 3) and declines at heavy usage (25+ hrs/week), suggesting a mild inverted-U relationship rather than a simple linear one. Traditional study hours ($r = 0.376$) and prompt-engineering skill remain far stronger academic predictors than AI usage volume.

Q2. Can we classify/predict Burnout_Risk_Level from study and AI habits?

Answer: Yes, with moderate success. A Random Forest classifier achieves 49.2% accuracy on the 3-class burnout outcome (vs. 33% baseline), and a binary logistic regression for High-burnout risk achieves a Pseudo R^2 of 0.20 with several strongly significant predictors (Section 9, 10). Weekly GenAI Hours and Perceived AI Dependency are, by a wide margin, the most important predictors in both models.

Q3. Does heavy AI reliance hurt long-term knowledge retention?

Answer: Yes, modestly. Weekly AI hours correlate negatively with skill retention ($r = -0.118$, $p < .001$), and the t-test confirms high-usage students retain significantly less than low-usage students (75.55 vs 76.04, $p < .001$). The effect is real but small. Interestingly, using a greater *diversity* of AI tools is associated with *higher* retention ($r = 0.197$) — suggesting it is not AI use per se that harms retention, but possibly narrow, repetitive reliance on a single tool.

Q4. How do student outcomes differ under different institutional AI policies?

Answer: Strict bans are associated with worse outcomes on both academic and well-being dimensions. Students under a Strict Ban show the smallest average GPA gain (+0.19 vs +0.21 under more permissive policies, Section 8) and the highest burnout risk (49% higher odds of High burnout than under an Actively Encouraged policy, Section 9). This is a striking finding: banning AI outright does not appear to protect students — if anything, it correlates with worse outcomes, possibly due to restricted-access stress, workaround behaviors, or policy-driven anxiety.

Q5. What are the links between AI dependency, exam anxiety, and burnout?

Answer: Strong and consistent. Perceived AI dependency correlates with exam anxiety at $r = 0.308$ ($p < .001$) — the strongest AI-related relationship in the entire dataset. In the logistic regression, dependency also independently predicts High burnout risk (OR = 1.13 per point, $p < .001$). Together with weekly AI hours (OR = 1.12), dependency is one of the two dominant drivers of burnout across every method tested — correlation, regression, and Random Forest.

Q6. Which disciplines/years benefit most or suffer most from AI adoption?

Answer: Differences by major are small; disciplines are not the main story. GPA change ranges only from +0.19 (Business) to +0.22 (STEM) across majors (Figure 1) — a narrow spread. The chi-square test confirms a statistically significant association between major and burnout risk ($\chi^2 = 376.27$, $p < .001$), but the Random Forest model assigns major only 1.3% feature importance, indicating this association is weak in practice once usage and dependency are accounted for. In short: **how** a student uses AI (intensity, dependency, prompting skill) matters far more than **which** major or year they belong to.

Conclusion & Recommendations

Overall Conclusion

Across every statistical method applied in this study — Pearson correlation, independent-samples t-tests, chi-square tests of independence, multiple linear regression, logistic regression, and Random Forest classification — a consistent, well-triangulated picture emerges:

AI usage intensity is a weak lever for academic grades but a strong lever for student well-being. Statistical significance was found almost everywhere (a natural consequence of $N = 50,000$), but effect sizes reveal two very different stories: small, often negligible effects on GPA and skill retention, versus moderate-to-strong, highly consistent effects on exam anxiety and burnout risk.

Recommendations

1. **Do not rely on outright bans.** Strict Ban policies are associated with both worse GPA outcomes and higher burnout risk. Institutions should consider structured, cited, or actively guided AI use rather than prohibition.
2. **Invest in prompt-engineering training.** Advanced prompters show a GPA advantage roughly 30× larger than the effect of raw AI usage hours — teaching effective AI use may be more valuable than restricting it.
3. **Monitor usage intensity and self-reported dependency as early-warning signals for burnout.** These two variables alone account for ~83% of predictive power in the Random Forest burnout model — a simple self-report dependency scale could serve as a low-cost screening tool.
4. **Encourage tool diversity over single-tool reliance.** Skill retention is higher among students using a greater variety of AI tools, suggesting engagement style matters more than total AI exposure.
5. **Continue traditional study-skills support.** Traditional study hours remain the single strongest predictor of GPA improvement ($r = 0.376$) — AI adoption should complement, not replace, foundational study habits.

Limitations

- All relationships are correlational; causal claims (e.g., "AI use causes burnout") cannot be confirmed from this observational dataset alone.
- Self-reported measures (perceived dependency, anxiety, prompt skill) are subject to self-report bias.
- The Random Forest model's 49.2% accuracy, while well above baseline, indicates substantial unexplained variance in burnout risk — other unmeasured factors (sleep, workload, personal circumstances) likely play a role.

Variable Dictionary

Variable	Type	Description
Student_ID	Integer	Unique numeric identifier for each student
Major_Category	Categorical	Field of study — STEM, Business, Humanities, Medical, Arts
Year_of_Study	Categorical	Academic standing — Freshman, Sophomore, Junior, Senior, Graduate
Pre_Semester_GPA	Float	GPA at the start of the semester
Post_Semester_GPA	Float	GPA at the end of the semester
Weekly_GenAI_Hours	Float	Average hours per week spent using Generative AI tools
Primary_Use_Case	Categorical	Main purpose for using AI
Prompt_Engineering_Skill	Categorical	Self-assessed proficiency in crafting effective prompts
Tool_Diversity	Integer	Number of distinct AI tools used (1-5)
Paid_Subscription	Boolean	Whether the student has a paid AI subscription
Traditional_Study_Hours	Float	Weekly hours devoted to traditional (non-AI) study methods
Perceived_AI_Dependency	Integer	Self-rated dependency on AI tools (1-10)
Institutional_Policy	Categorical	Institution's official stance on AI usage
Anxiety_Level_During_Exams	Integer	Self-reported exam anxiety score (1-10)
Skill_Retention_Score	Float	Score (0-100) measuring skill retention post-semester
Burnout_Risk_Level	Categorical	Assessed burnout risk — Low, Medium, High

Tools & Deliverables Summary

Deliverable	Format	Contents
Excel Summary Workbook	.xlsx	Raw data, descriptive stats, categorical frequencies, cross-tabulations (live formulas)
Stata Do-File	.do	Import, descriptive stats, frequency tables, contingency tables, ANOVA, correlation
Stata Cleaned Dataset	.dta	Encoded categorical variables, derived GPA_Change variable
Python Analysis Script	.py	Correlation, t-tests, linear/logistic regression, Random Forest classifier
R Analysis Script	.R	Equivalent correlation, t-test, regression, and classification workflow
PowerPoint Presentation	.pptx	12-slide summary of all statistical outputs and interpretations
Interactive Dashboards	.html	Three linked dashboards: Academic Performance, Well-Being & Burnout, Usage Patterns & Segments
This Report	.pdf	Comprehensive written analysis, hypotheses, all dashboard charts, and conclusions

Report generated 11 July 2026. All statistical tests use $\alpha = 0.05$. Analysis performed on ai_student_impact_dataset_1.csv (N = 50,000).